

MODULE B: BLENDER DESIGN BUILD

Step 0: Download Your Hardware Models

Before opening Blender, create a folder on your computer desktop named `Robot_CAD_Staging`.

1. Go to the [REV Robotics Technical Resources Page](#) [1]. Search for and download the **STEP (.step) files** for the *Core Hex Motor (REV-41-1300)*, the *6,000 RPM HD Hex Motor (REV-41-1291)*, and the *REV Control Hub (REV-31-1595)*.
2. Go to the [goBILDA ecosystem downloads catalog](#) and download the **STEP file** for the *1120 Series Structural Channels*.
3. **Convert to Blender Format:** Because Blender does not read raw CAD files easily, open the free conversion tool **FreeCAD**. Click *File* → *Import* to open each downloaded `.step` file. Click the part name in the left column so it turns green, then click *File* → *Export*. Choose **Wavefront OBJ (*.obj)** in the dropdown menu. Save them all into your folder. [3, 4, 5]

Step 1: Survive the Blender Screen (Your First 5 Minutes)

Open Blender. Click anywhere on the background to clear the landing splash screen.

- **The Default Box:** You will see a gray box sitting in the exact center of the grid canvas. Click on it so its edges highlight in bright orange, and press **Delete** on your keyboard to clear your canvas.
- **Look Around (Orbit):** Click and hold down the **Center Mouse Wheel (Scroll Wheel)** and move your mouse to rotate your view around the 3D room.
- **Move Side-to-Side (Pan):** Hold down the **Shift** key on your keyboard while holding down the **Center Mouse Wheel** and moving the mouse.
- **Zoom:** Scroll the wheel forward and backward.

✂ Step 2: Learn the Only 3 Keyboard Commands You Need

To build this entire robot piece-by-piece, you only need to use three keyboard shortcuts:

- **G Key (Move):** Click an object, press **G**, and slide your mouse to move it. If you want to move it perfectly straight along the red floor line, press **G**, then immediately press **X**. If you want to move it along the green floor line, press **G**, then **Y**.
- **R Key (Rotate):** Click an object, press **R**, and move your mouse to spin it. Pressing **R**, then **Z** spins it perfectly flat like a record player.
- **Duplicate:** Press **Ctrl + C** then **Ctrl + V** to copy any part instantly.

📁 Step 3: Loading Your Components & Laying Out the Drivetrain

1. Click **File** → **Import** → **Wavefront (.obj)** in the top menu bar. Navigate to your `Robot_CAD_Staging` folder, select your converted channel file, and import it.

2. Duplicate the channel to make 4 identical rails. Use **G (Move)** and **R (Rotate)** to connect them into a flat, perfect outer square frame measuring exactly **420mm wide x 420mm long** on the floor grid.
3. Click **File → Import → Wavefront (.obj)** again and import your `Core_Hex_Motor` mesh and wheel models.
4. Move **Core Hex Motor #1** into the rear-left corner cavity of your frame rails, and **Core Hex Motor #2** into the rear-right corner cavity. Snap your traction tires straight onto their output axle shafts.
5. Import your passive omni-wheel models. Slide one wheel unit into the **front-left corner corridor** of the bumper frame, and the second wheel unit into the **front-right corner corridor** to create your stable 4-wheel balance track.
6. Import the **REV Control Hub** and your single vertical **12V NiMH battery brick** model. Place them flat side-by-side in the center-rear internal bay floor.

✂ Step 4: Mounting the Overhung Saws & Cutting Your Nose Plate Slits

1. Import your triangular support tower block. Slide it forward along the green line (**G → Y**) until it sits right over the front bumper channel, **overhanging slightly past the main front edge line**.
2. Import a horizontal cylinder to act as your **12mm dead-shaft axle**, passing it straight through the tower's peak.
3. Duplicate your saw disc to make **two identical vertical saw wheels**. Place one saw disc on the left face of the triangular tower, and the second disc on its right face.
4. Add a flat plate shaped like a sloped front wedge ramp. Finish the main face in an aggressive **Glossy Red Material** and paint the inner frame rails **Matte Black**.
5. Select the red wedge face. Enter Edit Mode (**Tab Key**), select the **Knife Tool (Key: K)** or use rectangular loop cuts to slice two deep vertical window paths right where the saws sit. Slide the saws forward inside these tracks until their front teeth visibly project **40mm clear past the lowest leading edge of the red wedge**, completely removing any weapon reach bottlenecks.